



KSI
KUMARAGURU
SCHOOL OF
INNOVATION

24ADC001- BUSINESS INTELLIGENCE

HAMPHIRITHA BS – 24BAD403



2026-2027
LAB RECORD

**DEPARTMENT OF ARTIFICIAL
INTELLIGENCE AND DATA SCIENCE**

INDEX

Expt. No	Title of the Experiment	Page No	Total Marks	Faculty Signature
1.	Explore the interface and basic features of the BI tool			
2.	Load and visualize a sample dataset.			
3.	Import a dataset into the BI tool. And Cleanse data by handling missing values, outliers, and inconsistencies.			
4.	Transform data to suit BI reporting requirements and Design a dashboard with key performance indicators (KPIs).			
5.	Develop interactive dashboards for dynamic data exploration.			
6.	Integrate data from various sources for comprehensive analysis			
7.	Implement advanced chart types (treemaps, heatmaps, etc.).			
8.	Apply BI tools for forecasting and predictive analytics.			

Average marks

Faculty Signature

EXPT NO:1	EXPLORE THE INTERFACE AND BASIC FEATURES OF BI Tool
DATE: 10.07.2026	

PRE-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

1. What is the purpose of Business Intelligence (BI) tools like Tableau and Power BI?

Business Intelligence (BI) tools help users collect, analyze, visualize, and present data in the form of charts, graphs, and dashboards. They support better decision-making by converting raw data into meaningful insights.

2. List any three common data sources that Tableau and Power BI can connect to.

- Microsoft Excel
- CSV (Comma-Separated Values) files
- SQL Databases (e.g., MySQL, SQL Server)

3. What is a dashboard in BI, and how is it different from a single chart?

A dashboard is a collection of multiple visualizations, KPIs, filters, and reports displayed on a single screen for interactive analysis.

A single chart displays only one type of visualization, while a dashboard combines multiple charts to provide a complete overview of the data.

4. Define the following terms:

a) Visualization

Visualization is the graphical representation of data using charts, graphs, maps, or tables to make information easier to understand.

b) KPI (Key Performance Indicator)

A KPI is a measurable value used to evaluate how effectively a business or process is achieving its objectives.

c) Data Filter

A data filter allows users to display only the data that matches selected conditions or criteria.

5. Why is it important to clean and transform data before visualization?

Cleaning and transforming data removes errors, missing values, duplicates, and inconsistencies. This improves data quality and ensures accurate analysis and reliable visualizations.

6. What kind of chart would you use to analyze stock price trends over time?

A Line Chart is the most suitable chart because it clearly shows how stock prices change over time.

7. Explain the role of slicers in Power BI and filters in Tableau.

- Slicers (Power BI): Interactive controls that allow users to filter report data dynamically.
- Filters (Tableau): Used to display only selected data in worksheets or dashboards, making analysis more focused.

8. What does a calculated field or measure do in BI tools?

A calculated field or measure performs calculations on existing data to create new values, such as profit, percentage, or prediction error.

Example:

Prediction Error = $\text{ABS}(\text{Predicted Price} - \text{Close})$

9. What are the advantages of using interactive dashboards for financial data analysis?

- Provides real-time insights.
- Makes it easier to identify trends and patterns.
- Allows users to filter and analyze specific data.
- Supports faster and better decision-making.
- Combines multiple visualizations on one screen.

10. Which function would you use to find the absolute difference between predicted and actual stock prices?

Use the ABS() function.

Example:

$\text{ABS}(\text{Predicted Price} - \text{Close})$

This returns the absolute difference between the predicted price and the actual closing price.

IN-LAB EXERCISE:

Objective:

To understand the basic interface, data import process, and fundamental visualization capabilities of **Tableau** and **Power BI** using a stock prediction dataset, and to explore their usefulness in analyzing financial data.

Tools Required:

- Tableau Public / Tableau Desktop
- Power BI Desktop
- Sample Dataset (CSV/Excel) containing:
Date, Stock Name, Open, Close, High, Low, Volume, Predicted Price

Exercise Steps:

Part 1: Working with Tableau

1. **Launch Tableau** and connect to the stock prediction dataset.
2. **Explore the Interface:** ○ Understand components like **Data Pane**, **Sheets**, **Marks Card**, and **Dashboards**.
3. **Create Basic Visualizations:**
 - **Line Chart:** Show Close Price over time.
 - **Scatter Plot:** Plot Volume vs. Predicted Price.
 - **Highlight Table:** Display Predicted Price by stock.

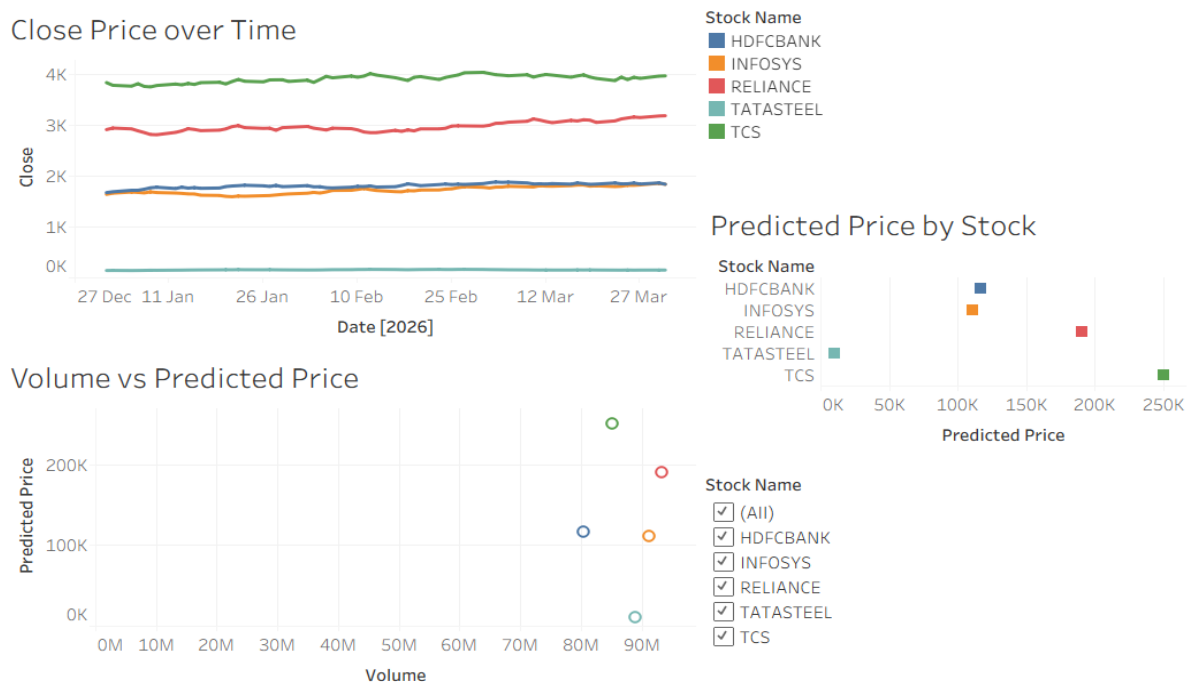
4. **Apply Filters:**
 - Add filters for Stock Name and Date Range.
5. **Create Calculated Field:** ○ Calculate prediction error as |Predicted - Close|.
6. **Design a Basic Dashboard:**
 - Combine your charts into one interactive dashboard.

Part 2: Working with Power BI

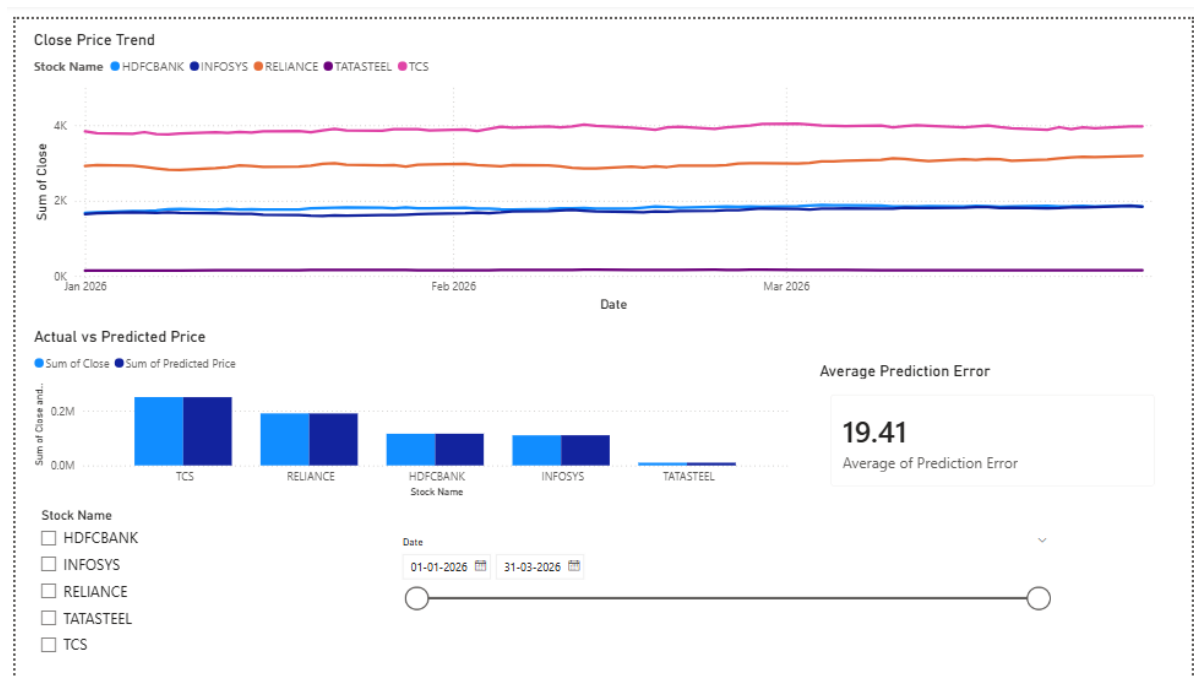
1. **Open Power BI Desktop** and import the same dataset.
2. **Explore the Interface:** ○ Familiarize with **Fields Pane**, **Visualizations Pane**, **Report View**, **Data View**.
3. **Create Basic Visualizations:**
 - **Line Chart:** Trend of Close Price over time.
 - **Bar Chart:** Compare Predicted Price vs. Close Price.
 - **Card (KPI):** Display average prediction error.
4. **Use Filters and Slicers:**
 - Filter data based on Stock Name and Date.
5. **Create a Simple Measure** using DAX:
 - Example: Prediction Error = ABS(Predicted - Close)
6. **Build a Report Page:**
 - Arrange visuals in a single interactive page.

OUTPUT :

Part 1: Working with Tableau



Part 2: Working with Power BI



POST-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

1. Describe your experience with the Tableau interface. What components did you interact with?

The Tableau interface was easy to understand and use. I interacted with the Data Pane, Sheets, Marks Card, Columns and Rows shelves, Filters, and Dashboard to create and organize visualizations.

2. How did you import the dataset into Tableau and Power BI?

In Tableau, I selected Connect → Microsoft Excel/CSV, chose the stock prediction dataset, and loaded it into the workbook. In Power BI, I clicked Home → Get Data → Excel, selected the same dataset, and loaded it into the report.

3. What types of visualizations did you create in both tools? List at least two per tool.

Tableau: Line Chart (Close Price over Time), Scatter Plot (Volume vs. Predicted Price), Highlight Table (Predicted Price by Stock).

Power BI: Line Chart (Close Price Trend), Bar Chart (Predicted Price vs. Close Price), Card (Average Prediction Error).

4. How did you calculate the prediction error, and how was it displayed in your dashboard?

I created a calculated field/measure using the formula $\text{Prediction Error} = \text{ABS}(\text{Predicted Price} - \text{Close Price})$. In Power BI, it was displayed using a Card (KPI) showing the average prediction error. In Tableau, it was created as a calculated field for analysis.

5. Which filtering or slicing techniques did you apply, and what insights did they reveal?

I applied Stock Name and Date filters in Tableau and Stock Name and Date slicers in Power BI. These helped analyze individual stock performance over different time periods and made the dashboard interactive.

6. Compare the process of creating a line chart in Tableau vs. Power BI. What differences did you notice?

In Tableau, I dragged Date to the Columns shelf and Close Price to the Rows shelf. In Power BI, I selected the Line Chart visual and assigned Date to the X-axis and Close Price to the Y-axis. Tableau uses drag-and-drop shelves, while Power BI uses a visual with field wells for X-axis, Y-axis, and Legend.

7. How did you organize the dashboard in Tableau and Power BI? What key elements did you include?

I arranged all visualizations on a single dashboard/report page. The dashboard included a Line Chart, Scatter Plot/Bar Chart, Highlight Table or KPI Card, Stock Name Filter/Slicer, and Date Filter/Slicer to make the dashboard interactive and easy to analyze.

8. Which tool (Tableau or Power BI) did you find more user-friendly for financial data visualization, and why?

I found Tableau more user-friendly because its drag-and-drop interface made it easier to create visualizations and dashboards quickly. Power BI was also powerful, especially for DAX calculations and report creation.

9. What challenges did you face while creating visualizations or dashboards?

Some challenges included choosing the correct chart type, managing filters and slicers, handling multiple data sources in Tableau, and creating calculated fields while ensuring the correct fields were used in visualizations.

10. How can the knowledge gained from this lab be applied to real-world financial data analysis?

The knowledge gained from this lab can be used to analyze stock market data, monitor financial performance, compare predicted and actual values, identify trends, and create interactive dashboards that support better business and investment decisions.

ASSESSMENT

Description	Max Marks	Marks Awarded
Pre Lab Exercise	5	
In Lab Exercise	10	
Post Lab Exercise	5	
Viva	10	
Total	30	
Faculty Signature		